the references dereferenced. All the browser needs to do is cross-reference the code in that downloaded file.

If you are already bundling your JavaScript code into one big package, all kinds of additional features become possible or desirable, such as minifying code. And once you've got a big bundling tool, all kinds of other tools attach to it. Walk far enough down that road, you get to webpack.

What import maps ask is, what if you didn't do any of that? What if you had a bunch of individual files on your server and told the browser which files correspond to which imports? Then you don't need to bundle everything in to a single package, which means you don't need the time or complexity overhead of a bundling tool.

In the distant past of, like, three years ago, this would have been impossible for two reasons. First, browsers did not support enough modern JavaScript to handle untranspiled code. In particular, browsers did not support `import` of ES6 modules natively until at least 2017 and in some cases 2019. Second, until HTTP2, the performance of serving lots of small files was significantly worse than serving one large file; that has changed, at least in theory.

And so, the import maps standard for declaring the relationship between import statements and code files was created. It's supported natively in some browsers, and handled with a lightweight shim in those browsers that don't handle them natively. (For more information on import maps themselves, you can go to the main site for the specification at https://github.com/WICG/import-maps.)

Even in an import maps world, however, you still need to manage dependencies and generate the code for the import maps that are to sent to the browser. Rails has the importmap-rails gem to manage those features.

What the importmap-rails gem does is allow you to specify your dependencies in a manifest file and then convert those dependencies to the HTML and JavaScript needed to send the actual import maps to the browser.

If you choose importmap-rails as your JS solution, it is installed as part of the